

Court Judgment Explainer — ROI Model (Justice Lens UK)

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1. ASSUMPTIONS — ROLES & RATES

Line item	Source / notes	Conservative	Normal
Annual salary — case desk staff (£)	Inferred: CharityJob 2026 UK charity median salary £36,000 (Normal); £30,000 admin/support-tier floor (Conservative). NOT Justice Lens's actual payroll.	£30,000	£36,000
Annual salary — senior reviewer (£)	Inferred: £42,000 (Cons.) / £48,000 (Normal) — senior/legally-literate reviewer tier. NOT confirmed.	£42,000	£48,000
Employment on-cost multiplier	Standard UK assumption: employer NI + pension + overhead ≈ +25% on top of salary.	1.25x	1.25x
Productive hours per FTE per year	≈45 working weeks (52 – ~5 weeks annual leave/bank holidays) × 37.5 hrs/week. This is ONE PERSON's available hours — used to convert salary to an hourly rate AND to convert annual workload into headcount (\$5).	1,700	1,700
Loaded hourly rate — case desk staff (£/hr)	Formula: salary × on-cost ÷ productive hours	£22.06	£26.47
Loaded hourly rate — senior reviewer (£/hr)	Formula: salary × on-cost ÷ productive hours	£30.88	£35.29
Organisational service coverage — distinct from the FTE hours above			
Org-wide operating weeks per year	This is how many weeks/year Justice Lens UK needs the case desk running (year-round service) — NOT one person's work-year. Individual holiday is already netted out via the 1,700-hr FTE figure above, so this stays 52 without double-counting leave.	52	52

2. IMPLEMENTATION / BUILD COST

Line item	Source / notes	Conservative	Normal
Team size (people)	Ross (PM) + 2 team members, per Stakeholder Interview 1 attendee list.	3	3
Build duration (working days)	2-week engagement, per project Gantt (10 working days).	10	10
Day rate per person (£/day)	Indicative commercial-equivalent rate — this is a coursework MVP with no fee actually charged. Conservative = UK median junior analyst day rate (£250, CalcKit 2026); Normal = junior-to-mid developer blended rate (£450, ContractorUK 2026 range £400-600).	£250	£450
Total implementation cost	Formula: team size × build days × day rate	£7,500	£13,500

3. CURRENT-STATE PROCESS — WEEKLY HOURS (org-wide total, all case-desk staff combined)

Line item	Source / notes	Conservative	Normal
Per-occurrence time — confirmed directly by Ben in interview (same both scenarios)			

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1. ASSUMPTIONS — ROLES & RATES

Hours per case-desk scan session	Confirmed: "about two hours".	2.00	2.00
Scan sessions per week	Confirmed: "three times per week".	3	3
Hours per case skim & log	Confirmed: "about 15 minutes".	0.25	0.25
Hours per full judgment read	Confirmed: "about half a day" — assumed 4 hrs; flag if org's working day differs.	4.00	4.00
Hours per briefing draft	Confirmed: "about three hours".	3.00	3.00
Hours per senior review	Confirmed: "around two hours" — Ben wants this checkpoint UNCHANGED, not compressed.	2.00	2.00
Hours per audience adaptation	Confirmed: "about an hour" — OUT OF SCOPE, stays manual, unaffected by this project.	1.00	1.00

Weekly volumes — NOT stakeholder-confirmed, see project_source_of_truth.md §4 correction

Raw judgments surfaced per week (scan volume)	Our team's own estimate stated in interview ("~30/day, or something like that") — Ben did not validate this figure. Conservative assumes it's overstated.	100	210
Candidates reaching skim/log (or Checkpoint 1 in new process)	Inferred from as-is narrative (scan → skim a narrower candidate set) — not a number Ben gave directly.	15	30
Cases reaching full read → draft → senior review → adapt	Inferred as a 'small number' per Ben's description — output volume assumed UNCHANGED between current and new process (same quality/quantity, per Ben's own framing).	3	6

Automation impact — calibrated to Ben's own two statements, not invented

New-process time factor (automatable steps only)	Conservative = Ben's floor ("divide time by three"): 1/3. Normal = Ben's stated triage target ("save 90% of the time") and his own bare-bones-win definition (1/10th the time): 1/10.	33.3%	10.0%
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Current-state weekly hours (org-wide total)

Scan (hrs/week)	Formula: hrs/session × sessions/week	6.00	6.00
Skim & log (hrs/week)	Formula: hrs/case × candidate volume	3.75	7.50
Full read (hrs/week)	Formula: hrs/case × read volume	12.00	24.00
Draft briefing (hrs/week)	Formula: hrs/case × read volume	9.00	18.00
Senior review (hrs/week)	Formula: hrs/case × read volume	6.00	12.00
Audience adaptation (hrs/week)	Formula: hrs/case × read volume — unaffected by project	3.00	6.00
TOTAL current weekly hours (all staff combined)	Sum of all steps above	39.75	73.50

4. NEW-STATE PROCESS — WEEKLY HOURS (with AI agent + human checkpoints)

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1. ASSUMPTIONS — ROLES & RATES

Line item	Source / notes	Conservative	Normal
Automated monitoring / spot-check (hrs/week)	Ingestion is fully automated (n8n poll of Find Case Law); residual = current scan × reduction factor	2.00	0.60
Checkpoint 1 — tick-list review (hrs/week)	NEW step, per Ben's design: human ticks AI-surfaced candidates instead of skimming cold. = current skim × reduction factor	1.25	0.75
Verify AI deep-dive extraction (hrs/week)	Replaces manual full read; human verifies AI-extracted facts/quotes/issues rather than reading cold. = current read × reduction factor	4.00	2.40
Edit AI-drafted briefing (hrs/week)	AI produces first draft; human edits rather than writing from scratch. = current draft × reduction factor	3.00	1.80
Senior review (hrs/week)	UNCHANGED — Ben explicitly wants this checkpoint preserved as the quality marker, not compressed.	6.00	12.00
Audience adaptation (hrs/week)	UNCHANGED — stays manual, out of scope for this project.	3.00	6.00
TOTAL new weekly hours (all staff combined)	Sum of all steps above	19.25	23.55

5. ANNUAL HOURS & HEADCOUNT REQUIRED (FTE)

Line item	Source / notes	Conservative	Normal
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Converts weekly org-wide hours into annual hours, then into how many full-time-equivalent people that actually represents — answers Ben's "lots and lots" of hours AND people directly, and separates the case-desk role from the senior-reviewer role.

Current state

Case-desk annual hours	(scan+skim+read+draft+adapt) × org-wide operating weeks	1,755.00	3,198.00
Case-desk FTE required	Formula: annual hours ÷ productive hours per FTE — this is headcount, not one person's total.	1.03 FTE	1.88 FTE
Senior-reviewer annual hours	Senior review hrs/week × org-wide operating weeks	312.00	624.00
Senior-reviewer FTE required	Formula: annual hours ÷ productive hours per FTE — realistically a part-time allocation, not a full role.	0.18 FTE	0.37 FTE

New state

Case-desk annual hours	(monitoring+checkpoint1+verify+draft+adapt) × org-wide operating weeks	689.00	600.60
Case-desk FTE required	Formula: annual hours ÷ productive hours per FTE	0.41 FTE	0.35 FTE
Senior-reviewer annual hours	Senior review hrs/week × org-wide operating weeks (unchanged)	312.00	624.00
Senior-reviewer FTE required	UNCHANGED from current state — Ben's checkpoint is preserved, not compressed.	0.18 FTE	0.37 FTE

6. ANNUAL LABOUR COST

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1. ASSUMPTIONS — ROLES & RATES

Line item	Source / notes	Conservative	Normal
Current-state annual labour cost	<i>(case-desk hrs × case-desk rate) + (senior hrs × senior rate)</i>	£48,349	£106,676
New-state annual labour cost	<i>(case-desk hrs × case-desk rate) + (senior hrs × senior rate)</i>	£24,834	£37,922

7. AI AGENT & TOOLING COSTS (new process only)

Line item	Source / notes	Conservative	Normal
AI inference cost			
First-pass triage model — input price (£/M tokens)	<i>Claude Haiku 4.5, \$1.00/M input; converted at ~£0.79/\$1 (approx FX, verify current rate before relying on this).</i>	£0.79	£0.79
First-pass triage model — output price (£/M tokens)	<i>Claude Haiku 4.5, \$5.00/M output; converted at ~£0.79/\$1.</i>	£3.95	£3.95
Tokens/case — triage input (title, metadata, excerpt)	<i>Estimate — not measured.</i>	1,500	1,500
Tokens/case — triage output (flag + short rationale)	<i>Estimate — not measured.</i>	150	150
Annual first-pass triage AI cost	<i>Formula: (in tokens×in price + out tokens×out price)/1,000,000 × scan volume × weeks</i>	£9.24	£19.41
Deep-dive extraction model — input price (£/M tokens)	<i>Claude Sonnet 5, introductory \$2.00/M input (through 31 Aug 2026, then \$3.00); converted at ~£0.79/\$1.</i>	£1.58	£1.58
Deep-dive extraction model — output price (£/M tokens)	<i>Claude Sonnet 5, introductory \$10.00/M output (then \$15.00); converted at ~£0.79/\$1.</i>	£7.90	£7.90
Tokens/case — deep-dive input (full judgment text)	<i>Estimate: ~5,000-word judgment × ~1.3 tokens/word + prompt overhead. Not measured — verify against real documents in Build.</i>	7,000	7,000
Tokens/case — deep-dive output (facts, issue, quotes, terms, draft)	<i>Estimate — not measured.</i>	1,200	1,200
Annual deep-dive extraction AI cost	<i>Formula: (in tokens×in price + out tokens×out price)/1,000,000 × checkpoint-1 volume × weeks</i>	£16.02	£32.04
Infrastructure & software			
AWS RDS hosting (£/year)	<i>db.t4g.micro compute ~\$11.68/mo (~£9.22/mo) + storage/backup/transfer headroom. Researched, not quoted.</i>	£180	£300
n8n hosting (£/year)	<i>Cloud Starter \$24/mo (~£19/mo, 2,500 executions) Conservative; Cloud Pro \$60/mo (~£47/mo, 10,000 executions) Normal.</i>	£228	£564
Tableau licensing (£/year)	<i>Creator seat \$75/user/mo annual (~£59/mo = £708/yr) Conservative, single seat; + 1 Viewer seat (~£12/mo=£144/yr) Normal.</i>	£708	£852
TOTAL AI + tooling annual cost	<i>Sum of AI inference + infrastructure + software</i>	£1,141	£1,767

8. ROI SUMMARY

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1. ASSUMPTIONS — ROLES & RATES			
Line item	Source / notes	Conservative	Normal
Current-state total annual cost	Formula: current-state annual labour cost	£48,349	£106,676
New-state total annual cost	Formula: new-state labour + AI + tooling	£25,975	£39,689
Annual savings	Formula: current total – new total	£22,373	£66,987
% cost reduction	Formula: savings ÷ current total	46.3%	62.8%
Payback period (months)	Formula: implementation cost ÷ (annual savings ÷ 12)	4.0 months	2.4 months
3-year net benefit	Formula: (annual savings × 3) – implementation cost	£59,620	£187,462

Key caveats

- Case volumes (raw scan, checkpoint-1 candidates, full-read output) are NOT stakeholder-confirmed — see project_source_of_truth.md §4. Replace with real figures once the MVP has run for a few weeks.
- Salary and loaded-rate assumptions are inferred from 2026 UK charity-sector salary survey data (CharityJob), not Justice Lens's actual payroll — ask for real figures if this model needs to go beyond an indicative estimate.
- The automation time-reduction factors (÷3 Conservative, ÷10 Normal) are deliberately anchored to Ben's own two statements in the interview, not invented — but they are still an assumption about how well the system will actually perform, not a measured result.
- AI token costs are estimated from typical judgment length, not measured against real documents — confirm during Build once real judgments are run through the pipeline.
- Senior review and audience adaptation are modelled as UNCHANGED in the new process by design, per Ben's explicit request to preserve that checkpoint — this is a deliberately conservative choice, not an oversight.
- Headcount (FTE) figures are a capacity calculation (annual hours ÷ productive hours per person), not a confirmed org chart — Justice Lens may split this work across more or fewer real people than the FTE math implies.
- Implementation cost is an indicative commercial-equivalent figure (3 people × 10 days × a researched UK day rate) — this is a coursework MVP with no fee actually charged; used here specifically to produce a meaningful payback period, not as a real invoice.
- 52 org-wide operating weeks is deliberately distinct from the 1,700-hour FTE figure: the organisation needs the case desk covered year-round, while any individual employee's holiday is already netted out of their productive-hours denominator, not the weeks figure — this avoids double-counting leave.